Realistic Visual Content Creation from In-The-Wild Images

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My research focuses on building machine learning algorithms for creating realistic visual content from in-the-wild images and in a self-supervised fashion. Most importantly, these algorithms do not require human-annotated, labeled training data. Such algorithms are going to have an enormous impact in not just creating beautiful visual content but also generating rich, diverse, and controllable training data for various computer vision tasks.

Recent state-of-the-art (SOTA) methods have shown evidence for the emergence of a neural rendering paradigm – a data-driven approach, enabling explicit or implicit control of scene properties such as illumination, camera parameters, pose, geometry, appearance, and semantic structure. However, these SOTA approaches cannot generalize and scale. They require 3D geometry and material model of objects, and scene's illumination map or several views of the scene. That means an enormous collection of training data with their labeled material properties that are difficult to get and annotate. Therefore, most of these methods rely on computer graphics generated synthetic data and cannot generalize to real scenes. In contrast, my goal is to develop a general-purpose, image-based neural renderer that does not require any labeled training data and that can learn from in-the-wild and easily available comprehensive collection of image data alone.

In the last year, I have developed methods: **DIPR** and **IlluminatOR** for realistic image-based object insertions; **Relighter** for realistic image-based scene relighting; and **ViewGen** for synthesizing high fidelity textured 3D object models. DIPR and IlluminatOR render in-the-wild objects that are cut from one or more images and pasted into another. Both these methods produce insertions that are convincing with realistic lighting effects *only* from cut-and-paste images. With our proposed methods, we neither require the geometry and material model of the inserted object nor the target scene's illumination map. Our Relighter is the first scene relighting method that requires no data marked up in any form. Relighter generates multiple diverse and realistic relighted images of general scenes when trained on a standard dataset with no scenes in two different light settings. ViewGen generates textured 3D models of objects from a 2D single image input without ever seeing 3D object data while training. ViewGen fixes the problem of view generalization that the previous approaches struggle with while generating realistic rendered object novel views from inferred textured 3D models. For all these methods, neither we have ground-truth data nor we collect labeled training data. They are all self-supervised.

Next, my goal is to develop a method – **Spotlight** and extend DIPR and IlluminatOR to videos where the goal is to insert new objects that are cut from one video and transplanted into another. For Spotlight, I will develop a method for learning representations of the scene from a video that combines ideas from multi-view stereo for extracting dense 3D representation and current learning-based methods for rendering objects into scenes that are temporally and spatially realistic, and consistent with the target scene's lighting. This is challenging because it requires us to be aware of and track the locale of an inserted object in 4D and infer spatial dynamics of other objects surrounding it to better model: (a) occlusions at different time steps and (b) inserted object's appearance with spatially and time-varying lighting of the scene. After this, I will develop a unified rendering framework – **Lightformer**, a general-purpose image-based relighting system. Lightformer will be controllable and complete all our light editing tasks like inserting new objects into a static or dynamic scene and entirely change the scene's illumination given a target style. Further, we will extend Lightformer to reason about human scene interaction and generate novel views of the scenes (say from Hollywood movies), with realistic lighting effects.